

DAY 14 — TABLE-NEIGHBORHOOD RETRIEVAL + CONTEXT EXPANSION

✓ GOAL

Move from:

row-level retrieval ✗

to:

table-aware retrieval ✓

so engineering relationships stay intact.

✓ WHAT DAY 14 FIXES

✓ average/max comparison

✓ front/rear comparison

✓ CDT/GHD/HSD understanding

✓ multi-row engineering reasoning

✓ weak isolated chunk retrieval

✓ incomplete context

✓ CORE IDEA

If one row is retrieved:

Front pad peak temp | Average

automatically retrieve:

neighbor rows

from same table

✓ NEW ARCHITECTURE

Semantic Retrieval

↓

Neighbor Expansion ✓

↓

Deduplication 

↓

Structured Context 

↓

Qwen2.5 

TEMPLATE STRUCTURE

app/

└─ retrieval/

| └─ neighbor_retriever.py 

|

└─ agent/

| └─ ask.py  MODIFY

STEP 1 — CREATE NEIGHBOR RETRIEVER

 **app/retrieval/neighbor_retriever.py**

```
from app.storage.duckdb_store import (
```

```
    conn
```

```
)
```

```
def get_neighbor_chunks(
```

```
    table_id,
```

```
    row_index,
```

```
    window=2
```

```
):
```

```
min_row = max(0, row_index - window)
```

```
max_row = row_index + window
```

```
query = f"""
```

```
SELECT content
```

```
FROM engineering_chunks
```

```
WHERE table_id = '{table_id}'
```

```
AND CAST(
```

```
    regexp_extract(
```

```
        chunk_id,
```

```
        'row_(\d+)',
```

```
        1
```

```
    ) AS INTEGER
```

```
)
```

```
BETWEEN {min_row} AND {max_row}
```

```
"""
```

```
rows = conn.execute(query).fetchall()
```

```
return [r[0] for r in rows]
```

✅ STEP 2 — MODIFY `hybrid_search.py`

 `app/query/hybrid_search.py`

Add import:

```
from app.retrieval.neighbor_retriever import (  
    get_neighbor_chunks  
)
```

✅ ADD AFTER RERANKING

Replace:

```
return list(reranked[:5])
```

WITH:

```
expanded_results = []
```

```
for r in reranked[:5]:
```

```
    expanded_results.append(r)
```

```
    metadata = r["metadata"]
```

```
    table_id = metadata["table_id"]
```

```
    row_index = metadata.get(  
        "row_index",
```

```
        0
```

```
    )
```

```
    neighbors = get_neighbor_chunks(  
        table_id,
```

```
        table_id=table_id,
        row_index=row_index,
        window=2

    )

    for n in neighbors:

        expanded_results.append({
            "document": n,
            "metadata": metadata,
            "source": "neighbor"
        })

    return expanded_results
```

✅ STEP 3 — ENSURE ROW INDEX EXISTS

 **app/embeddings/chunker.py**

Make sure metadata contains:

```
"row_index": idx
```

✅ FINAL CHUNK FORMAT

```
chunks.append({

    "id": f"{table_id}row{idx}",

    "text": text,

    "metadata": {
```

```
"page": page,  
  
"table_id": table_id,  
  
"row_index": idx,  
  
"source": "table"  
}  
})
```

✅ STEP 4 — DEDUPLICATION STILL USED

Your existing:

 **context_deduplicator.py**

already works.

No change needed ✅

✅ STEP 5 — RUN

python main.py

✅ NOW TEST

Compare front pad and rear liner peak temperature at average

✅ BEFORE DAY 14

Retriever got only:

Front pad | Average

❌ incomplete context

✓ AFTER DAY 14

Retriever automatically expands:

Front pad | Average

Front pad | Maximum

Rear liner | Average

Rear liner | Maximum

✓ complete engineering neighborhood

✓ EXPECTED RESPONSE

Front pad average temperatures:

104 | 104 | 82

Rear liner average temperatures:

94 | 98 | 72

Observation:

Front pad temperatures are consistently higher.

Possible higher thermal loading observed

in front braking system.

✓ WHY THIS IS THE CORRECT SOLUTION

Engineering meaning exists in:

table structure ✓

NOT isolated rows.

✓ IMPORTANT

This is:

- ✓ generic
 - ✓ layout-aware
 - ✓ no hardcoding
 - ✓ works for any engineering table
 - ✓ retrieval-native
-

✓ CURRENT SYSTEM

PDF

↓

Extraction

↓

Normalization

↓

Filtering

↓

Embeddings

↓

ChromaDB

↓

Hybrid Retrieval

↓

Neighbor Expansion ✓

↓

Deduplication ✓

↓

Context Intelligence ✓

↓

Qwen2.5 ✓

✅ **YOU ARE NOW HERE**

✅ DAY 14 COMPLETE

Next:

DAY 15

- multi-document reasoning
- report comparison
- cross-PDF retrieval
- engineering workflow planning
- autonomous tool selection